

Reviewer Pack — Minimal Symplectic Form Rank vs BP_ReadTwice Circuit Depth

Entry adf1cb892fac · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Symplectic Form Rank vs BP_ReadTwice Circuit Depth

Entry ID: adf1cb892fac **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-22 12:45:20 UTC

1. Conjecture

Field A (mathematical branch): Symplectic Geometry **Field B** (complexity object): Complexity Theory: Branching Program Read-Twice Complexity

Statement:

{'text': 'For any read-twice branching program P, the minimal rank of its associated symplectic form is $\Theta(\log^2 \text{size}(P))$ ', 'counterexample': 'There exists a counterexample where the minimal rank of the symplectic form for a given read-twice branching program is less than $\Theta(\log^2 \text{size}(P))$ '}

Rationale (proposer's reasoning):

{'text': 'Symplectic geometry offers a structured way to encode the complexity of computations, potentially providing deeper insights into the limitations of BP_read_twice circuits. The minimal symplectic form rank could serve as a non-trivial invariant that distinguishes between read-once and read-twice complexities.'}

Taxonomy category: BP_READTWICE (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): c9cae12a3ab858b0

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for at least 80% of generated read-twice branching programs with $n \leq 40$, the ratio of the minimal rank to $\log^2(n)$ falls within $\pm 30\%$ of its expected value ($\Theta(\log^2 \text{size}(P))$).

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 6 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "symplectic geometry" AND "branching program read-twice complexity" - "minimal rank" symplectic form AND $\theta(\log^2)$ size(P) - "counterexample" minimal rank symplectic form < $\theta(\log^2)$ size(P)

Top relevant hits considered: - [http://arxiv.org/abs/1908.01012v3] Model comparison from LIGO-Virgo data on GW170817's binary components and consequences for the merger remnant - [http://arxiv.org/abs/2011.00217v1] Searches for 25 rare and forbidden decays of D^+ and D_s^+ mesons - [http://arxiv.org/abs/2012.05143v2] First observation of the decay $B_s^0 \rightarrow K^- \mu^+ \nu_\mu$ and a measurement of $|V_{ub}|/|V_{cb}|$ - [http://arxiv.org/abs/2007.11292v2] First observation of the decay $\Lambda_b^0 \rightarrow \eta_c(1S) p K^-$ - [http://arxiv.org/abs/2407.14261v2] Study of charmonium production via the decay to $p\bar{p}$ at $\sqrt{s} = 13\text{TeV}$ - [http://arxiv.org/abs/2103.07349v2] Measurement of the prompt-production cross-section ratio $\sigma(\chi_{c2})/\sigma(\chi_{c1})$ in $p\text{Pb}$ collisions at $\sqrt{s_{NN}} =$

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, $\leq 90\text{s}$ wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_read_twice_circuit(n):
        if n == 1:
            return [[0]]
        else:
            left = generate_read_twice_circuit(n // 2)
            right = generate_read_twice_circuit(n - n // 2)
            circuit = []
            for i in range(len(left)):
                circuit.append([left[i][0] + right[i][0]])
            return circuit

    def symplectic_form_rank(circuit):
        n = len(circuit)
        if n == 1:
            return 1
        m = n // 2
        A = [[0 for _ in range(n)] for _ in range(m)]
        for i in range(m):
```

```

        for j in range(n):
            A[i][j] = circuit[i][0][j]
    C = [[A[i][j] - A[i+m//2][j+m//2] for j in range(n)] for i in range(m)]
    rank = 0
    for row in C:
        if any(row):
            rank += 1
    return rank

n_values = [5, 10, 15, 20, 30, 40]
results = []

for n in n_values:
    circuit = generate_read_twice_circuit(n)
    rank = symplectic_form_rank(circuit)
    expected_rank = math.log2(n) ** 2
    ratio = rank / expected_rank if expected_rank != 0 else float('inf')
    results.append({
        "n": n,
        "rank": rank,
        "expected_rank": expected_rank,
        "ratio": ratio
    })

mean_ratio = sum(result["ratio"] for result in results) / len(results)
std_ratio = math.sqrt(sum((result["ratio"] - mean_ratio) ** 2 for result in results) / len(results))
support_fraction = sum(1 for result in results if abs(result["ratio"] - 1) < 0.3) / len(results)

return {
    "metric_name": "Ratio of Symplectic Form Rank to Expected Rank",
    "metric_value": mean_ratio,
    "instances_tested": len(results),
    "conjecture_holds": support_fraction >= 0.8,
    "counterexample": "" if support_fraction >= 0.8 else f"n={results[0]['n']}, rank={results[0]['rank']}"
}

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_ratio = sum(result["metric_value"] for result in results) / len(results)
    std_ratio = math.sqrt(sum((result["metric_value"] - mean_ratio) ** 2 for result in results) / len(results))
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_ratio} std={std_ratio} support_fraction={support_fraction}")
    else:
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\n n={results[0]['n']}, rank={results[0]['rank']}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
, 'counterexample': 'n=5, rank=1, expected_rank=5.391350077827255'}  
TRIAL: {'metric_name': 'Ratio of Symplectic Form Rank to Expected Rank', 'metric_value': 0.0786652101.  
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```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Safety rail: critic_challenge_falsified | original: The test results indicate that for at least one instance (n=5), the minimal rank of the symplectic form is less than half of the expected value, which | next: Investigate further to understand why the test crashed and whether it was a fluke or indicative of a deeper issue with the conjecture. If the crash is due to an error in the implementation, fix the code and retest.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13403
2	preregistration	ollama_remote	glm4:latest	0	0	9677
3	novelty	ollama_remote	glm4:latest	0	0	8662
4	novelty	ollama_remote	glm4:latest	0	0	9474
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9703
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9694
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	22935
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11809
9	judge	ollama_remote	glm4:latest	0	0	60428

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 155784 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/adf1cb892fac.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/adf1cb892fac.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/adf1cb892fac.tar.gz` (if generated)